

An Analysis of Image to image translation

Image to image translation

Image to image translation -- Part 1

Alternative algorithms A range of other methods exist for solving simple as well as higher order MRFs. They include Maximization of Posterior Marginal, Multi-scale MAP estimation, Multiple Resolution segmentation and more. Apart from likelihood estimates, graph-cut using maximum flow and other highly constrained graph based methods exist for solving MRFs.

Image to image translation -- Part 2

The Laplacian operator is employed such that the partial derivatives are derived from a specific equation. The second partial derivative of $f(x, y)$ with respect to x and y are given by:

Image to image translation -- Part 3

Content-based image retrieval Machine vision Medical imaging, and imaging studies in biomedical research, including volume rendered images from computed tomography, magnetic resonance imaging, as well as volume electron microscopy techniques such as FIB-SEM. Locate tumors and other pathologies Measure tissue volumes Diagnosis, study of anatomical structure Surgery planning Virtual surgery simulation Intra-surgery navigation Radiotherapy Digital Pathology and Histopathology. Nuclei instance segmentation in variously stained whole slide images (WSIs) refers to the automatic delineation of individual cells' nuclei borders. This task is a specialized subproblem of instance segmentation, with high practical importance in biomedical research and diagnostics (e.g. cell counting, morphometric feature extraction, tumor grading, prognostic biomarker extraction). The modern approach to the problem is the use of deep learning architectures developed to encompass the main challenges of separating overlapping or fused nuclei. The availability of few humanly-annotated datasets poses another challenge to the problem. Object detection Pedestrian detection Face detection Brake light detection Locate objects in satellite images (roads, forests, crops, etc.) Recognition Tasks Face recognition Fingerprint recognition Iris recognition Prohibited Item at Airport security checkpoints Traffic control systems Video surveillance Video object co-segmentation and action localization Several general-purpose algorithms and techniques have been developed for image segmentation. To be useful, these techniques must typically be combined with a domain's specific knowledge in order to effectively solve the domain's segmentation problems.

Image to image translation -- Part 4

Semi-automatic segmentation In one kind of segmentation, the user outlines the region of interest with the mouse clicks and algorithms are applied so that the path that best fits the edge of the image is shown. Techniques like SIOX, Livewire, Intelligent Scissors or IT-SNAPS are used in this kind of segmentation. In an alternative kind of semi-automatic segmentation, the algorithms return a spatial-taxon (i.e. foreground, object-group, object or object-part) selected by the user or designated via prior probabilities.

Image to image translation -- Part 5

$$\nabla^2 f(x, y) = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}$$

Image to image translation -- Part 6

Model-based segmentation The central assumption of model-based approaches is that the structures of interest have a tendency towards a particular shape. Therefore, one can seek a probabilistic model that characterizes the shape and its variation. When segmenting an image, constraints can be imposed using this model as a prior. Such a task may involve (i) registration of the training examples to a common pose, (ii) probabilistic representation of the variation of the registered samples, and (iii) statistical inference between the model and the image. Other important methods in the literature for model-based segmentation include active shape models and active appearance models.

Image to image translation -- Part 7

Markov random fields The application of Markov random fields (MRF) for images was suggested in early 1984 by Geman and Geman. Their strong mathematical foundation and ability to provide a global optimum even when defined on local features proved to be the foundation for novel research in the domain of image analysis, de-noising and segmentation. MRFs are completely characterized by their prior probability distributions, marginal probability distributions, cliques, smoothing constraint as well as criterion for updating values. The criterion for image segmentation using MRFs is restated as finding the labelling scheme which has maximum probability for a given set of features. The broad categories of image segmentation using MRFs are supervised and unsupervised segmentation.

Image to image translation -- Part 8

$$\frac{\partial^2 f(x, y)}{\partial y^2} = f(x, y+1) + f(x, y-1) - 2f(x, y)$$

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Image to image translation -- Part 9

Simulated annealing (SA) Derived as an analogue of annealing in metallurgy, simulated annealing (SA) uses change in pixel label over iterations and estimates the difference in energy of each newly formed graph to the initial data. If the newly formed graph is more profitable, in terms of low energy cost, given by:

Image to image translation -- Part 10

$$\alpha (1 - \delta(\mathbf{i} - \mathbf{i}_{\text{initial}})) + \beta \sum_{q \in N(i)} (1 - \delta(\mathbf{i}, \mathbf{q}(i))) . \quad \{\displaystyle \alpha (1-\delta(\ell_{\mathbf{i}}-\ell_{\{\text{initial}\}})) + \beta \sum_{q \in N(i)} (1-\delta(\ell_{\mathbf{i}}, \ell_{\mathbf{q}(i)}))\}.$$

Image to image translation -- Part 11

$$\operatorname{argmin}_{u, K} \gamma |K| + \mu \int K C |\nabla u|^2 dx + \int (u - f)^2 dx . \quad \{\displaystyle \operatorname{argmin}_{u, K} \gamma |K| + \mu \int K C |\nabla u|^2 dx + \int (u - f)^2 dx . \}$$

Image to image translation -- Part 12

Optimization algorithms Each optimization algorithm is an adaptation of models from a variety of fields and they are set apart by their unique cost functions. The common trait of cost functions is to penalize change in pixel value as well as difference in pixel label when compared to labels of neighboring pixels.

Image to image translation -- Part 13

Image segmentation and primal sketch There have been numerous research works in this area, out of which a few have now reached a state where they can be applied either with interactive manual intervention (usually with application to medical imaging) or fully automatically. The following is a brief overview of some of the main research ideas that current approaches are based upon. The nesting structure that Witkin described is, however, specific for one-dimensional signals and does not trivially transfer to higher-dimensional images. Nevertheless, this general idea has inspired several other authors to investigate coarse-to-fine schemes for image segmentation. Koenderink proposed to study how iso-intensity contours evolve over scales and this approach was investigated in more detail by Lifshitz and Pizer. Unfortunately, however, the intensity of image features changes over scales, which implies that it is hard to trace coarse-scale image features to finer scales using iso-intensity information. Lindeberg studied the problem of linking local extrema and saddle points over scales, and proposed an image representation called the scale-space primal sketch which makes explicit the relations between structures at different scales, and also makes explicit which image features are stable

over large ranges of scale including locally appropriate scales for those. Bergholm proposed to detect edges at coarse scales in scale-space and then trace them back to finer scales with manual choice of both the coarse detection scale and the fine localization scale. Gauch and Pizer studied the complementary problem of ridges and valleys at multiple scales and developed a tool for interactive image segmentation based on multi-scale watersheds. The use of multi-scale watershed with application to the gradient map has also been investigated by Olsen and Nielsen and been carried over to clinical use by Dam. Vincken et al. proposed a hyperstack for defining probabilistic relations between image structures at different scales. The use of stable image structures over scales has been furthered by Ahuja and his co-workers into a fully automated system. A fully automatic brain segmentation algorithm based on closely related ideas of multi-scale watersheds has been presented by Undeman and Lindeberg and been extensively tested in brain databases. These ideas for multi-scale image segmentation by linking image structures over scales have also been picked up by Florack and Kuijper. Bijaoui and Rué associate structures detected in scale-space above a minimum noise threshold into an object tree which spans multiple scales and corresponds to a kind of feature in the original signal. Extracted features are accurately reconstructed using an iterative conjugate gradient matrix method.